

AutoKAN: A Federated Lightweight Anomaly Detection Framework for Securing Constrained IoT Healthcare Diabetes Monitoring Systems

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Abstract—The adoption of Internet of Things (IoT) technology in healthcare has significantly enhanced patient care by improving both the efficiency and cost-effectiveness of healthcare delivery systems. Specifically, for diabetic patients, IoT facilitates continuous health monitoring, enabling healthcare professionals to track patients' conditions in real time and detect potential complications at an early stage. However, the growing number of IoT devices has also introduced security concerns, particularly in terms of cyber threats that exploit resource limitations such as power and memory. Especially in the healthcare sector, where data is highly sensitive and distributed in nature, privacy is one of the most critical aspects. A promising privacy-preserving technology, known as federated learning, addresses these challenges effectively. This research presents a lightweight federated anomaly detection framework tailored for constrained environments, including small-scale IoT devices. We conceptualize IoT devices as essential components within the edge continuum and leverage anomaly detection models to enhance security. Our anomaly detection model is built on an autoencoder architecture; however, rather than relying on conventional multilayer perceptron (MLP) networks, we utilize Kolmogorov–Arnold Networks (KAN) with an adaptive threshold to minimize parameter complexity and enable real-time deployment. Additionally, the federated learning mechanism is applied to ensure the privacy of patient data is safeguarded. Experimental results indicate that our model achieves an accuracy exceeding 99.5% and a precision of 100%, outperforming traditional autoencoders while utilizing 50% fewer parameters

and achieving twice the speed in training and inference time. This demonstrates the superiority of our approach in anomaly detection, particularly in fields where precision is the most important metric, such as medical applications.

Index Terms—Federated learning, smart healthcare, IoT, anomaly detection, Kolmogorov–Arnold networks, diabetes monitoring.

SYMBOL DEFINITIONS

- L : Total number of layers in a KAN network.
- z_0 : Input to the KAN, $z_0 \in \mathbb{R}^{d_0}$.
- Ψ_l : Function matrix corresponding to the l -th layer in KAN.
- $\psi_{l,j,i}$: Activation function linking layer l to layer $l+1$, where $l = 0, 1, 2, \dots, L-1$; $i = 1, 2, \dots, d_l$; $j = 1, 2, \dots, d_{l+1}$.
- d_l : Number of nodes in layer l .
- d_{l+1} : Number of nodes in layer $l+1$.
- $z_{l,i}$: Input to $\psi_{l,j,i}$.
- $\tilde{z}_{l,j,i}$: Output of $\psi_{l,j,i}$.
- $z_{(l+1),j}$: Activation value of the $(l+1, j)$ neuron.
- α_s : A learnable parameter in the activation function.
- α_r : A learnable parameter in the activation function, initialized according to Xavier initialization.
- $\mathcal{S}(z)$: A learnable function computed using B-spline basis functions.
- w_l^t : AutoKAN model at round t , where $l \in \{1, 2, 3, \dots, L\}$ and L is the number of devices.
- Z : Input to the Encoder network, $(z_1, z_2, z_3, \dots, z_n)$.
- K : Encoder network.
- h : Output of the Encoder network, $K(Z)$.
- K' : Decoder network.
- Z' : Output of the Decoder network, $K'(h)$.
- $L_{AutoKAN}$: Loss function, calculated using Mean Squared Error (MSE).
- g_t : Gradient at time step t .
- θ : Learnable parameters $\{\alpha_s, \alpha_r, c\}$.
- m_t : Moving average of gradients.
- v_t : Moving average of squared gradients.
- β_1, β_2 : Hyperparameters for momentum and variance decay.
- η : Learning rate.
- ϵ : Small constant for numerical stability.
- n_l : Number of samples on device l .
- N : Total dataset size across selected devices.
- E_i : Reconstruction error for Z_i .

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- λ_{max} : Maximum reconstruction error.
- λ_{min} : Minimum reconstruction error.
- E_i^{test} : Reconstruction error for test data T_i .

I. INTRODUCTION

DIABETES is a chronic health condition that arises when the pancreas fails to produce sufficient insulin or when the body is unable to use insulin effectively. The World Health Organization (WHO) reported that in 2021, an estimated 830 million individuals were affected by diabetes, leading to approximately 2 million deaths. Furthermore, elevated blood glucose levels were linked to nearly 11% of deaths associated with cardiovascular diseases [1].

Advancements in IoT-based biomedical devices are playing a crucial role in both self-management for diabetic patients and supporting healthcare organizations responsible for patient care [2]. IoT healthcare sensors and smart devices collect a variety of health parameters, including blood pressure, glucose levels, heart rate, and electrocardiogram (ECG) readings. The gathered data is then transmitted to a healthcare server for storage and analysis via wireless networks, such as WLANs, PANs, or WMANs. In recent years, the Ambient Assisted Living (AAL) methodology has been introduced to improve care for elderly diabetic patients. This approach enables caregivers to monitor blood sugar and glucose levels in real time using smart devices. Additionally, IoT-enabled solutions, such as ambulance dispatch systems, ensure immediate medical assistance in emergency situations. Anomaly detection in IoT healthcare refers to identifying observations, events, or data points that deviate significantly from normal patterns, distinguishing them from the rest of the sensor dataset. Identifying and handling anomalies is crucial because they can impact the decision-making process of medical professionals and directly affect patient well-being. Furthermore, detecting anomalies improves data quality, contributing to the integrity and reliability of data, which is essential not only for training machine learning models but also for accurate patient diagnosis. The main cause of unusual events is sensor inaccuracies, which can lead to anomalies for various reasons, such as low latency, cyberattacks, or hardware failures. Additionally, anomalies can arise in resource-constrained environments.

In the healthcare domain, Abououf et al. [3] introduced an interpretable method for anomaly detection, utilizing an autoencoder to rapidly identify anomalous events, while a deep neural network handles classification. To enhance explainability, the authors incorporated KernelSHAP. Likewise, Aguilar et al. [4] integrated a decision tree with an autoencoder to enhance the interpretability of the autoencoder's outputs, offering domain experts a more intuitive explanation. Although autoencoders offer a promising approach for anomaly detection, their performance still needs improvement [5], [6], [7], [8], particularly in IoT systems, where millions of data points are generated daily. Even a 1% error rate can translate into a significant number of misdetections, making accuracy crucial in such environments.

In IoT systems, edge devices often have limited computational resources, making the development of lightweight anomaly

detection models essential. Fan et al. [9] introduced LUAD, a lightweight anomaly detection framework that integrates a Temporal Convolutional Network with a Variational autoencoder. Their approach significantly reduces the number of parameters—nearly ten times fewer than the OmniAnomaly model—while maintaining comparable accuracy. Additionally, several studies have also focused on designing lightweight anomaly detection models for edge devices [10], [11], [12]. However, many of these works primarily demonstrate their effectiveness in high-computing environments and often rely on a limited set of evaluation metrics, which may not fully validate their performance in real-world edge computing scenarios. On the other hand, privacy is also an important aspect of healthcare, and concerns about data security often hinder the sharing of healthcare information, thereby limiting opportunities for improvement and innovation in the field.

This research aims to address the following questions:

- 1) What level of accuracy and precision can a lightweight anomaly detection framework based on Kolmogorov–Arnold Networks (KANs) achieve in IoT-based healthcare monitoring systems?
- 2) How does the proposed framework perform in resource-constrained environments compared to traditional autoencoder-based methods?
- 3) How does integrating federated learning into the AutoKAN framework support the preservation of patient data privacy while maintaining effective anomaly detection in decentralized healthcare settings?

To address the challenges mentioned above, this work proposed a novel deep federated learning anomaly autoencoder-based detection in IoT healthcare system, focusing on lightweight, performance, privacy, and precision detection. Inspired by the latest existing research, an autoencoder-based model has been developed due to the advantage of Kolmogorov–Arnold Networks [13] in performance. The reconstructed error is calculated and used as the anomaly score with an adaptive threshold to ensure the scalability and model accuracy, especially in an uncertainty environment like IoT. Therefore, a federated learning mechanism is proposed to aggregate local data while ensuring patient privacy. Hence, the contribution of this work can be written as follows:

- 1) Develop the AutoKAN federated anomaly detection framework for healthcare IoT, ensuring high accuracy, efficiency, privacy and robustness in detecting anomalies within a controlled environment. The model is designed to adapt to real-world healthcare scenarios, where reliability and privacy are critical.
- 2) Simulate a resource-constrained computing environment to evaluate the efficiency and scalability of the proposed model. This includes testing under limited computational resources to assess the feasibility of deploying AutoKAN in edge and IoT-based healthcare systems. The results showcase the model's proficiency in terms of execution time, inference speed, and overall computational complexity, ensuring its practicality for real-time applications.
- 3) Enhance detection reliability by minimizing false positives and false negatives through an adaptive

thresholding mechanism. This approach dynamically adjusts detection thresholds based on contextual factors, improving accuracy while ensuring scalability. The proposed method contributes to more reliable anomaly detection, making it suitable for large-scale deployments in diverse healthcare environments.

The structure of this paper is organized as follows: Section II reviews related work. Section III describes the methodology for stress prediction and the experimental design. Further details on the experimental setup are provided in Section IV. Section V evaluates the performance of the proposed model and includes a discussion. The study's limitations and potential future research directions are outlined in Section VI-C. Finally, Section VII summarizes the key findings of this paper.

II. RELATED WORK

Anomaly detection in IoT, particularly in healthcare, has emerged as a critical area of research due to the increasing connectivity of medical devices and the associated security risks. This section provides a detailed exploration of anomaly detection techniques, frameworks, and models tailored for IoT healthcare applications, drawing insights from recent studies. Additionally, a Comparison of related work is detailed in Table I.

A. Anomaly Detection in IoT

Anomaly events may be well-reconstructed but difficult to distinguish. Zeng et al. [26] applied particle swarm optimization to optimize hyperparameters for an adversarial VAE for anomaly detection. Their method achieves high performance in various domains, such as secure water treatment, water distribution, the Mars Science Laboratory, and power systems. Katbi et al. [27] propose a novel IoT anomaly detection system that uses a one-class approach with a deep learning architecture. Their methodology involves integrating a Deep SVDD objective function with an adversarial interpolated network structure to achieve effective grouping of normal data points in the latent space. Using a multi-head graph attention network (MD-GAT) to capture feature correlations and a temporal convolution network to learn temporal dependencies, Tang et al. [28] proposed a supervised contrastive learning-based spatiotemporal variational autoencoder (SC-STVAE). Vijai and Sivakumar [14] propose a novel anomaly detection approach for industrial applications using a BiLSTM-Variational Autoencoder (BiLSTM-VAE) model with a dynamic loss function. The model aims to overcome the limitations of traditional anomaly detection methods, such as scalability issues, high false alarm rates, and reliance on skilled expertise.

Focusing on energy-efficient models by reducing anomaly-irrelevant sensory data transmitted within the network, Guo et al. [15] developed EGNN, an anomaly detection method utilizing a graph neural network. Nonetheless, this approach lacks generalization and scalability. Utilizing a bias-trained artificial intelligence neural network to formulate a set of rules at each node of a tree-based algorithm,

Sivapalan et al. [16] developed an interpretable rule-mining solution for detecting anomalies in real-time ECG signals. Additionally, the model achieves over 50% power savings in sensor consumption by transmitting only anomaly events. However, its performance is not highly comparable, reaching only approximately 85-90% accuracy. Achieving 100% effectiveness in malware detection for Linux-based IoT, Breitenbacher et al. [17] proposed HADES-IoT, a host-based anomaly detection system that not only ensures high accuracy but also maintains low overhead, making it suitable for high-performance demanding tasks. Xu et al. [29] proposed an unsupervised framework to jointly optimize the graph contrastive learning module and the network reconstruction module for high-precision anomaly detection in networks of consumer electronic devices. Their approach shows superior results on three public benchmark datasets. AlMahadin et al. [30] applied the deep learning technique GRU to detect anomalies such as DDoS attacks in Vehicular Ad-hoc Networks (VANETs). Similarly, Nissar et al. [31] employed a variational autoencoder with various optimization objectives, including divergence, KL-divergence, and reconstruction error, using the AGE-MOEA and R-NSGA-III algorithms. Zhao et al. [32], using a convolutional encoder and a deconvolutional decoder—each leveraging the parallelism of stacked Advanced Dilated Causal Convolutional (ADCC) blocks—demonstrated that their approach outperforms traditional methods in anomaly detection within satellite orbit environments. Zukai et al. [33] introduced a meta-learning framework that demonstrated superior performance compared to other algorithms across multiple datasets, including USTL-EHMS-2020, IoTID20, and WUSTL-IIOT-2021, attaining an accuracy of 99%. However, ensemble methods may not be well-suited for IoT-based environments where real-time applications demand rapid processing. Fang et al. [18] developed an anomaly detection model tailored for medical IoT systems, utilizing rough set theory and fuzzy core vector machines. Despite its novel approach, the model's accuracy remains approximately 92%. Albattah and Rassam [19] employed a convolutional long short-term memory (ConvLSTM) network to detect anomalies in wireless body area networks. Experimental findings validated the model's effectiveness, reporting a 98% F1-score and 99% accuracy, surpassing traditional CNN and LSTM-based models. Likewise, Shaikh et al. [34] achieved an impressive 99.78% accuracy in intrusion detection within the Internet of Medical Things (IoMT) by leveraging a hybrid approach combining CNN and LSTM architectures. Building on LSTM-based methodologies, Wang et al. [35] concentrated on constructing a distributed anomaly detection system that ensures both high precision and low latency. In a similar vein, Hasniuj et al. [36] proposed a robust framework capable of identifying anomalous behaviors in IoT devices while also implementing mechanisms to restrict and mitigate their effects on interconnected systems. Their solution incorporates a lightweight machine learning model designed for accurate packet-level anomaly detection. Additionally, Balaji et al. [20] explored an ensemble-based strategy by integrating SqueezeNet and NasNet architectures to train a

TABLE I
COMPARISON OF RELATED WORK IN ANOMALY DETECTION FOR IOT HEALTHCARE SYSTEMS

Author(s)	Method	Key Features	Application	Limitations
Menatalla et al. [3]	Interpretable Anomaly Detection	Autoencoder for rapid anomaly identification + Deep Neural Network for classification + KernelSHAP for explainability.	Healthcare monitoring systems	Interpretable, rapid identification.
Vijai et al. [14]	BiLSTM-Variational Autoencoder (BiLSTM-VAE)	Dynamic loss function to overcome traditional method limitations.	Industrial applications, IoT environment	Aims to reduce false alarms and scalability issues.
Guo et al. [15]	EGNN (Energy-Efficient Graph Neural Network)	Reduces anomaly-irrelevant sensory data transmitted within the network.	IoT multivariate time series data	Energy-efficient, but lacks generalization and scalability.
Gawsalyan et al. [16]	Interpretable Rule-Mining Solution	Bias-trained AI neural network to formulate rules at each node of a tree-based algorithm.	Real-time ECG signals in IoT edge sensors	Interpretable, ~50% power savings, but approx. 85-90% accuracy.
Zukai et al. [17]	Meta-learning framework (Meta-IDS)	Meta-learning approach.	IoMT network, USTL-EHMS-2020, IoTID20, WUSTL-IIOT-2021	Superior performance, 99% accuracy. Ensemble methods may not suit real-time IoT.
Fang et al. [18]	Anomaly detection model	Rough set theory + fuzzy core vector machines.	Medical IoT systems	Novel approach, approx. 92% accuracy.
Shaikh et al. [19]	Hybrid CNN-LSTM approach (RelNet)	Combines CNN and LSTM architectures.	Internet of Medical Things (IoMT) intrusion detection	Impressive 99.78% accuracy.
Balaji et al. [20]	Ensemble-based strategy	Integrates SqueezeNet and NasNet architectures.	Healthcare-related IoT infrastructures (intrusion detection)	Deep learning model for intrusion detection.
Elayan et al. [21]	Deep Federated Learning	Federated learning for monitoring and analyzing healthcare data.	Skin disease detection	Sustainable healthcare data analysis.
Akter et al. [22]	Three-fold Federated Edge Aggregator	Enhances security against privacy attacks.	Healthcare systems	Approx. 90% accuracy on various datasets.
Siva et al. [23]	Federated learning with NFTs	Patients store data in NFT marketplace; addresses FL challenges with Polyak-averaging.	Smart healthcare	Privacy-preserving medical-data-sharing.
Lin et al. [24]	Deep Federated Learning	Graph convolutional networks to gain user trust.	IoT healthcare systems	Privacy-aware access control.
Li et al. [25]	Federated supervised learning model	Addresses data heterogeneity.	Heterogeneous IoT healthcare systems	Outperforms state-of-the-art methods in mitigating imbalanced class distributions.
Our Approach	AutoKAN	Lightweight federated anomaly detection based on KAN autoencoder with adaptive threshold.	Constrained IoT healthcare (diabetes monitoring)	99.3% accuracy, 100% precision, 50% fewer parameters, 2x speed of traditional autoencoders.

deep learning model aimed at intrusion detection within healthcare-related IoT infrastructures.

B. Federated Learning In IoT Healthcare System

With the rise of privacy concerns and the growing number of data breaches across industries, the need for technologies that safeguard data privacy while maintaining accuracy and sustainability has become crucial. Federated learning is gaining attention as a promising solution to address these challenges. Elayan et al. [21] introduced a deep federated learning approach for monitoring and analyzing healthcare data to detect skin diseases. Meanwhile, Akter et al. [22] developed a three-fold federated edge aggregator to enhance the security of healthcare systems against privacy attacks. The effectiveness of their model was evaluated using the MNIST, CIFAR-10, STL-10, and COVID-19 chest X-ray datasets, achieving an accuracy of approximately 90%. Utilizing Non-Fungible Tokens (NFTs) with federated learning, Sai et al. [23] encouraged patients to store data in the NFT marketplace and

addressed some challenges in federated learning using the Polyak-averaging method. To enhance privacy-aware access control in IoT healthcare systems, Lin et al. [24] proposed a deep federated learning approach that utilizes graph convolutional networks to gain user trust by obtaining users' influences. Li et al. [25] developed a federated supervised learning model focused on solving the data heterogeneity problem. This approach outperforms state-of-the-art methods such as SCFed, CBAFed, and FedCD by providing a suitable strategy to mitigate imbalanced class distributions.

C. Kolmogorov-Arnold Networks

Nowadays, in addition to the popular MLP network, various new architectures are emerging to explore novel structures with better interpretability and higher performance. Abd Elaziz et al. [13] introduced the Kolmogorov-Arnold Network (KAN), drawing inspiration from the Kolmogorov-Arnold representation theorem. The Kolmogorov-Arnold Network has

recently garnered significant attention for its strong interpretability and exceptional performance. Seydi et al. [37] compared various KAN variants, including Fully Connected KAN, Chebyshev-KAN, Radial Basis Function KAN, and Naïve-Fourier-KAN, for hyperspectral change detection. Their experiments demonstrated that Chebyshev-KAN achieved the best performance for this task. In another study, Jiang et al. [38] employed the Kolmogorov–Arnold Network for short-term load forecasting in power systems. Their results demonstrated the effectiveness of the proposed model, as the RMSE score outperformed other methods, such as XGBoost and MLP. Liu et al. [39] utilized Kolmogorov–Arnold Networks for semi-supervised impedance inversion. Through experimental comparisons of acoustic impedance fitting results, they showed that KAN and Convolutional KAN exhibit stronger fitting capabilities for long-term acoustic impedance data than traditional linear layers and CNNs. To address the challenges related to limited computing and memory resources, Song et al. [40] propose an adaptive wavelet transform Kolmogorov–Arnold network approach for hybrid control devices using motor imagery electroencephalography. Do et al. [41] evaluated and compared multiple KAN variants, such as Original-KAN, Fast-KAN, Jacobi-KAN, DeepKAN, and Chebyshev-KAN, for botnet classification. Their findings indicated that the Original-KAN achieved the best performance on the N-BaIoT, IoT23, and IoT-BotNet datasets. The application of KAN shows its potential for better performance in various tasks. This motivates us to conduct research with KAN for our work.

III. METHODOLOGY

A. KAN Layer

KAN emerges as a promising alternative to Multi-Layer Perceptrons (MLPs), delivering enhanced performance and faster computation. The primary difference between KAN and MLP lies in their activation functions: while MLPs rely on fixed activation functions, KAN utilizes learnable activation functions on the edges, which are subsequently aggregated at the nodes.

A KAN can be structured as a composition of multiple KAN layers. Let $\mathcal{L}$ denote the total number of layers in a KAN network, with the input represented as $\mathbf{z}_0 \in \mathbb{R}^{d_0}$. The transformation of KAN can be expressed as:

$$\mathcal{K}(\mathbf{z}) = (\Psi_{\mathcal{L}-1} \circ \Psi_{\mathcal{L}-2} \circ \dots \circ \Psi_0)(\mathbf{z}) \quad (1)$$

where Ψ_l represents the function matrix corresponding to the l -th layer in KAN. More specifically, we define $\psi_{l,j,i}$ with:

$$l = 0, 1, 2, \dots, \mathcal{L} - 1; \quad i = 1, 2, \dots, d_l; \quad j = 1, 2, \dots, d_{l+1} \quad (2)$$

where ψ denotes the activation function linking layer l to layer $l + 1$. Here, d_l and d_{l+1} represent the number of nodes in layers l and $l + 1$, respectively. Let $z_{l,i}$ be the input to $\psi_{l,j,i}$ and $\tilde{z}_{l,j,i}$ be the corresponding output. The activation value of the $(l + 1, j)$ neuron is computed as:

$$z_{(l+1),j} = \sum_{i=1}^{d_l} \tilde{z}_{l,j,i} = \sum_{i=1}^{d_l} \psi_{l,j,i}(z_{l,i}) \quad (3)$$

for $j = 1, \dots, d_{l+1}$. This can be rewritten as:

$$\mathbf{z}_{l+1} = \Psi_l \mathbf{z}_l \quad (4)$$

where:

$$\Psi_l = \begin{bmatrix} \psi_{l,1,1} & \psi_{l,1,2} & \psi_{l,1,3} & \dots & \psi_{l,1,d_l} \\ \psi_{l,2,1} & \psi_{l,2,2} & \psi_{l,2,3} & \dots & \psi_{l,2,d_l} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \psi_{l,d_{l+1},1} & \psi_{l,d_{l+1},2} & \psi_{l,d_{l+1},3} & \dots & \psi_{l,d_{l+1},d_l} \end{bmatrix} \quad (5)$$

The activation function is computed as:

$$\psi(z) = \alpha_r b(z) + \alpha_s \mathcal{S}(z) \quad (6)$$

where:

$$b(z) = \frac{z}{1 + e^{-z}} \quad (7)$$

and $\mathcal{S}(z)$ is computed as, with c_i are the learnable variable and $\mathcal{B}_i(z)$ are B-spline basis functions:

$$\sum_i c_i \mathcal{B}_i(z) \quad (8)$$

In the initial stage, $\alpha_s = 1$, α_r is initialized according to the Xavier initialization and $\mathcal{S}(z) \approx 0$.

A smaller KAN network can achieve accuracy comparable to or even surpass that of a larger MLP network by learning the activation functions directly. This is especially beneficial for developing lightweight models on devices with limited computational resources.

B. AutoKAN Architecture

Based on the KAN layer, we design the proposed AutoKAN, whose structure is illustrated in Figure 1. The AutoKAN model, denoted as w_l^t at round t , where $l \in \{1, 2, 3, \dots, L\}$ and L is the number of devices in the federated network, consists of two main components: the encoder and the decoder. Each component is built using a stack of KAN layers. After training for E local epochs, the weights are sent to the server to update the global model.

We denote the encoder network as K , with input $Z = (z_1, z_2, z_3, \dots, z_n)$. By applying equation (1), we obtain:

$$K(Z) = (\Phi_{L-1} \circ \Phi_{L-2} \circ \dots \circ \Phi_0)Z = h \quad (9)$$

Similarly, the encoder network is denoted as K' , with input $h = K(Z)$, with P denoted as the total number of layers in Decoder network, leading to:

$$K'(h) = (\Phi_{P-1} \circ \Phi_{P-2} \circ \dots \circ \Phi_0)h = Z' \quad (10)$$

The loss function is calculated using the Mean Squared Error (MSE):

$$L_{\text{AutoKAN}} = \frac{1}{N} \sum (Z' - Z)^2 \quad (11)$$

The learnable parameters in KAN are updated using the Adam optimizer. The optimization process can be described as follows:

$$g_t = \nabla_{\theta} J(\theta_t), \quad \text{where } \theta = \{\alpha_s, \alpha_r, c\} \quad (12)$$

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t \quad (13)$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2 \quad (14)$$

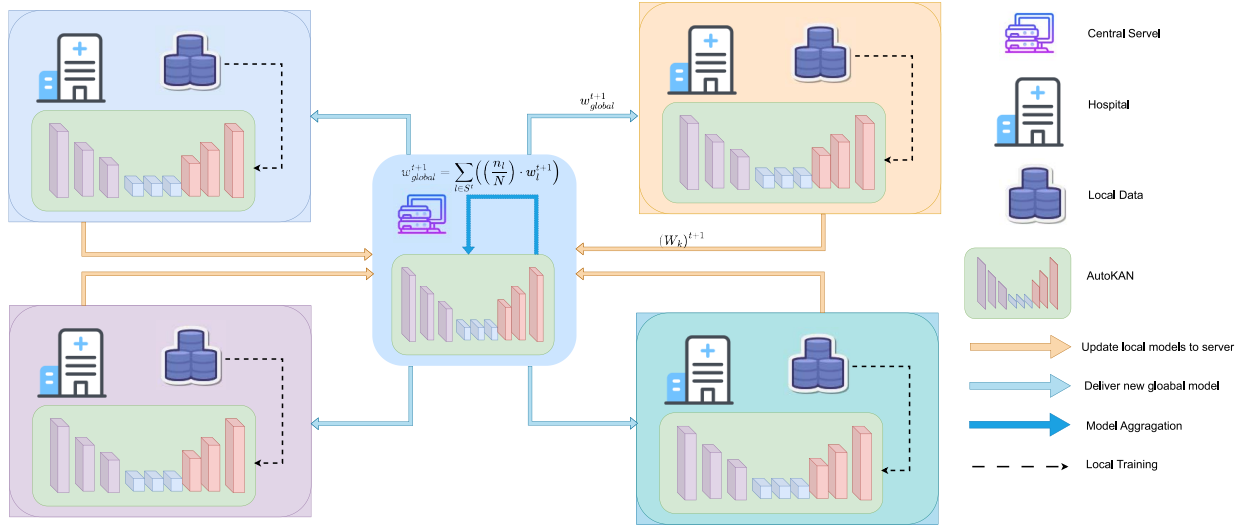


Fig. 1. Federated mechanism for anomaly detection for privacy-preserving IoT diabetes healthcare system. This illustrates the flow of information and model updates within the federated learning framework.

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}, \quad \hat{v}_t = \frac{v_t}{1 - \beta_2^t} \quad (15)$$

$$\alpha_s^{(t+1)} = \alpha_s^{(t)} - \frac{\eta}{\sqrt{\hat{v}_t} + \epsilon} \hat{m}_t \quad (16)$$

$$\alpha_r^{(t+1)} = \alpha_r^{(t)} - \frac{\eta}{\sqrt{\hat{v}_t} + \epsilon} \hat{m}_t \quad (17)$$

$$c^{(t+1)} = c^{(t)} - \frac{\eta}{\sqrt{\hat{v}_t} + \epsilon} \hat{m}_t \quad (18)$$

where η represents the learning rate, β_1 and β_2 serve as hyperparameters that influence momentum and variance decay, and ϵ is a small constant ensuring numerical stability by preventing division by zero. The gradient at time step t is denoted as g_t , while m_t and v_t correspond to the moving averages of gradients and squared gradients, respectively. Additionally, t signifies the current iteration step.

This step-by-step optimization ensures efficient parameter updates, allowing AutoKAN to learn feature representations effectively.

C. Federated Autoencoder Kolmogorov-Arnold Networks With Adaptive Threshold for Anomaly Detection

After training, the server aggregates updates using Federated Averaging mechanism.

$$w_{\text{global}}^{t+1} = \sum_{l \in S^t} \frac{n_l}{N} w_l^{t+1} \quad (19)$$

where n_l is the number of samples on device l and $N = \sum_{l \in S^t} n_l$ is the total dataset size across selected devices. The updated global model is then sent back to all devices, and this process repeats for T rounds until convergence. The final model w_{global}^T is deployed for inference. Details process is provided in Algorithm 1.

The process flow of testing and calculating adaptive threshold is illustrated in Algorithm 2. The normal data is fed into AutoKAN to calculate the reconstruction error. Note that the input data consists only of normal data. Our aim is to

Algorithm 1 Federated Learning Framework for Local Model Aggregation (FedAvg)

- 1: **Input:** Initial global model w_{global}^0 , number of participating devices L , local training epochs E , learning rate η , and total communication rounds T .
- 2: **Output:** Optimized global model w_{global}^T
- 3: **Server Initialization:**
- 4: The global model w_{global}^0 is initialized
- 5: The initialized model is distributed to all devices $l \in L$
- 6: **for** $t = 1$ to T **do**
- 7: Server selects a subset of devices $S^t \subseteq L$
- 8: **for each device** $l \in S^t$ **concurrently do**
- 9: Retrieve the current global model w_{global}^t
- 10: Train a local model on device l over E epochs, following equations (12)–(18)
- 11: Transmit the updated local model w_l^{t+1} back to the server
- 12: **end for**
- 13: The server applies FedAvg to aggregate the models:
- 14: $w_{\text{global}}^{t+1} = \sum_{l \in S^t} \frac{n_l}{N} w_l^{t+1}$
- 15: The updated global model w_{global}^{t+1} is then shared with all devices
- 16: **end for**
- 17: **Return** the final trained global model w_{global}^T

intentionally overfit the model to the training data so that techniques such as dropout and batch normalization are not required. After training, the reconstruction error E can be used to extract the threshold for normal data as follows:

$$\lambda_{\max} = \max(E), \quad \lambda_{\min} = \min(E)$$

During the testing phase, any value exceeding or falling below this boundary is considered an anomaly.

Algorithm 2 AutoKAN Anomaly Detection With Adaptive Threshold at the Local l^{th} Device in Round t

- 1: **Input:** Normal data Z , Test data T , Local model w_l^t
- 2: Compute the reconstruction error:

$$E_i = w_l^t(Z_i) - Z_i \quad \forall Z_i \in Z \quad (20)$$

- 3: $E \leftarrow E \cup \{E_i\}$
- 4: Compute anomaly thresholds:

$$\lambda_{\max} = \max(E), \quad \lambda_{\min} = \min(E) \quad (21)$$

- 5: **Test Phase:**
- 6: **for** each $T_i \in T$ **do**
- 7: Compute the reconstruction error:

$$E_i^{\text{test}} = \frac{1}{N} \sum (w_l^t(T_i) - T_i)^2 \quad (22)$$

- 8: **if** $E_i^{\text{test}} > \lambda_{\max}$ or $E_i^{\text{test}} < \lambda_{\min}$ **then**
- 9: T_i is classified as an **Anomaly**
- 10: **else**
- 11: T_i is classified as **Normal**
- 12: **end if**
- 13: **end for**
- 14: **Output:** Classification result (Anomaly or Normal)

IV. EXPERIMENT DESIGN

A. Dataset

The dataset simulates an IoT-based environment, specifically an Intensive Care Unit (ICU), populated with nine patient monitoring devices. The detailed specifications of the sensors integrated within this setup are illustrated in Figure 3. To rigorously evaluate the system's security and resilience, four distinct types of cyberattacks were meticulously simulated: Message Queuing Telemetry Transport (MQTT) Publish Flood, MQTT Authentication Bypass Exploit, MQTT Malicious Packet Injection, and Constrained Application Protocol (CoAP) Replay Exploit. The simulation was conducted using IoT-Flock [42], a specialized and open-source tool designed explicitly for modeling and generating traffic in diverse IoT environments. IoT-Flock was carefully configured to closely emulate real-world ICU equipment and operational dynamics, incorporating actual data profiles and time profiles derived from the analysis of real-time value ranges of ICU devices. During the simulation, network traffic, encompassing both normal and attack scenarios, was captured using Wireshark, a widely-used network protocol analyzer, to log both packet headers and payloads. Subsequently, a custom Python script was employed to extract relevant network-layer and application-layer features from the captured traffic, facilitating a detailed and focused analysis. The dataset, originally created and openly published by Hussain et al. [43], comprises 80,126 normal traffic samples and 76,810 attack traffic samples, meticulously labeled as 0 (normal) and 1 (attack), respectively. In our experiment, we allocated 70% of the data for training and 30% for testing to ensure robust model development and evaluation. However, given that our approach necessitates only normal data for training, we exclusively

utilized normal traffic from the 70% training set, resulting in a total of 56,044 normal traffic samples specifically used for model training.

B. System and Model Configuration

The experimental setup involved a high-performance computing system featuring an Intel Core i7-13700 processor (24 cores), 32GB of RAM, and an NVIDIA GeForce RTX 4060 GPU. The implementation was performed in Python 3.10.15, leveraging the PyOD library [44] to execute anomaly detection algorithms for benchmarking against the proposed method.

To further evaluate the model's efficiency in real-world deployment scenarios, we simulated a resource-constrained environment comparable to a Raspberry Pi 5, featuring 2GB RAM and a quad-core processor. This setup was used to assess the feasibility of deploying the proposed model in low-power edge computing devices.

The model was trained using Mean Squared Error (MSE) Loss, with the Adam optimizer and a total of 30 training epochs, learning rate $\alpha = 0.001$. This configuration ensures stable learning while minimizing reconstruction errors, making the model suitable for anomaly detection in both high-end and low-resource environments.

C. Evaluation Metrics

To comprehensively assess the performance of the proposed model, we consider multiple evaluation metrics.

1) Accuracy:

$$\text{Accuracy} = \frac{P_t + N_t}{P_t + N_t + P_f + N_f} \quad (23)$$

where P_t , N_t , P_f , and N_f represent true positives, true negatives, false positives, and false negatives, respectively.

2) Precision:

$$\text{Precision} = \frac{P_t}{P_t + P_f} \quad (24)$$

3) Recall (Sensitivity):

$$\text{Recall} = \frac{P_t}{P_t + N_f} \quad (25)$$

4) F1-Score:

$$F_1 = \frac{2PR}{P + R} \quad (26)$$

5) **Memory Usage:** Represents the amount of RAM required by the model during inference, measured in megabytes (MB) or gigabytes (GB).

6) **Floating Point Operations per Second (FLOPs):** Indicates the total number of floating-point calculations performed by the model, reflecting computational efficiency.

7) **Number of Parameters:** Represents the total learnable weights in the model, influencing both computational cost and generalization ability.

8) **Running Time:** Measures the time taken by the model to process whole test input samples, evaluated in seconds (s).

9) **Traffic Per Second:** Determines the number of data points or samples processed by the model per second.

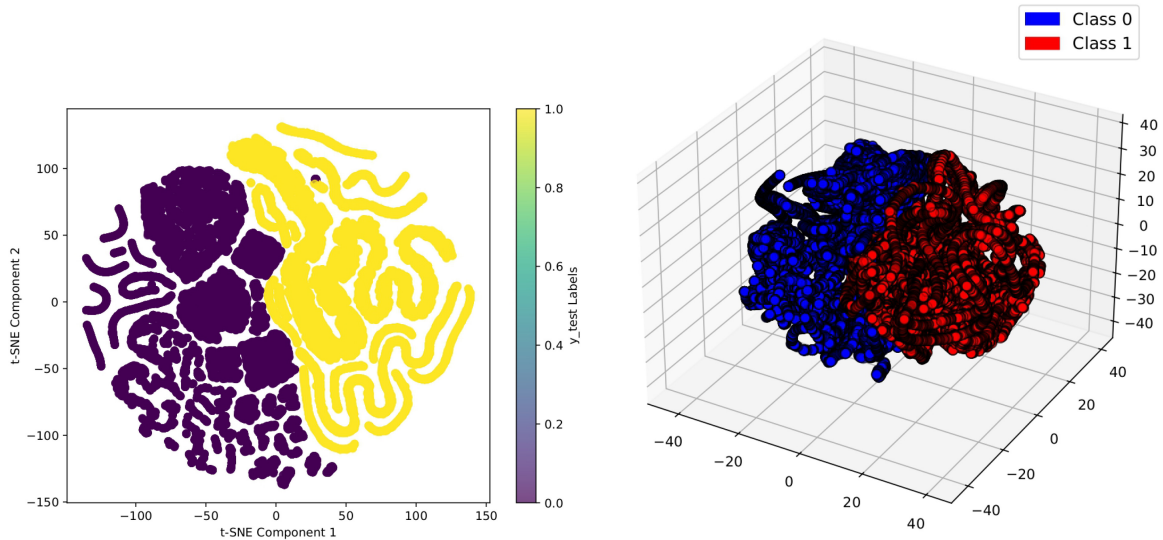


Fig. 2. Two-dimensional (left) and three-dimensional (right) t-SNE embeddings of the dataset, revealing distinct clusters corresponding to two different classes.

Sensor Name	Functionality	Measured Data
Remote Electrocardiogram (ECG)	Assesses heart electrical and muscular activity	Pulse Rate (0–200 bpm)
Infusion Pump	Delivers controlled doses of nutrients and medications to patients	Dose (10–100 mL)
Pulse Oximeter (SpO2)	Monitors blood oxygen saturation levels	Oxygen saturation (35–100%)
Nasal/Mouth Airflow Sensor	Captures respiratory rate through airflow measurement	Respiratory rate (0–60 peaks/min)
Blood Pressure Monitor	Measures systolic and diastolic pressure levels	Blood Pressure (0–300 mm Hg)
Glucometer	Determines blood glucose concentration	Glucose (10–150 mg/dL)
Body Temperature Sensor	Monitors body temperature fluctuations	Temperature (0–120°F)
Electromyography (EMG) Sensor	Detects electrical activity produced by muscles	Muscle contractions (0–60 cpm)
Galvanic Skin Response (GSR) Sensor	Measures electrical conductance of the skin	Conductance (0–20 μ S)

Fig. 3. Overview of Patient Monitoring Sensors.

These metrics collectively provide a comprehensive evaluation of the model’s accuracy, efficiency, and computational feasibility in different deployment scenarios.

V. EXPERIMENT RESULT

A. Comparison With State-of-the-Art Anomaly Detection Techniques

The results are presented in Table II, where it is evident that AutoKAN outperformed more than 28 methods, including deep learning, unsupervised learning, and statistical approaches, demonstrating its superior performance across various anomaly detection techniques. AutoKAN achieved an accuracy of 99.11% and a precision of 100%, indicating that the model produced no false positives. The high F1-score of 99.08% further highlights the model’s proficiency in the medical field, where minimizing errors is crucial. Once again, deep learning models clearly showcase their effectiveness in tasks such as anomaly detection, particularly when large amounts of data are available. For instance, deep learning-based models, such as autoencoder and AnoGAN, exhibit significantly better performance compared to other categories, such as Proximity-Based and Outlier Ensemble methods. This can be attributed to the increasing complexity of modern networks and IoT environments, where traditional approaches fail to capture intricate patterns. Deep learning models, on the other hand, are better equipped to learn from complex data distributions.

However, not all deep learning approaches perform equally well. Models such as DeepSVDD, MO_GAAL, and VAE exhibit relatively lower performance, suggesting that certain architectures may not be well-suited for specific anomaly detection tasks.

Figure 4 illustrates the training loss of a model over 30 epochs, showing a general trend of decreasing loss as training progresses. The loss decreases significantly in the initial epochs (0–20), indicating rapid learning, and then plateaus after epoch 20, suggesting convergence with diminishing returns from further training. While the low final loss suggests successful training, the plateauing raises the possibility of overfitting, which is the goal we are trying to achieve. This overfitting with normal training data aims to increase the model’s sensitivity to anomalies. Figure 5 reveals that the model performs exceptionally well in correctly classifying normal events, with 24,082 out of 24,082 instances correctly predicted. However, there is a small number of misclassifications in anomaly detection. Specifically, 420 anomaly events were incorrectly classified as normal. This indicates that while the model has a very low false positive rate, it exhibits a small but non-zero false negative rate.

Another reason that contributes to the superiority of the proposed method over other approaches is the adaptive thresholding strategy. Instead of using a fixed threshold, we define dynamic boundaries by taking the maximum and minimum values from the normal distribution. This allows the model

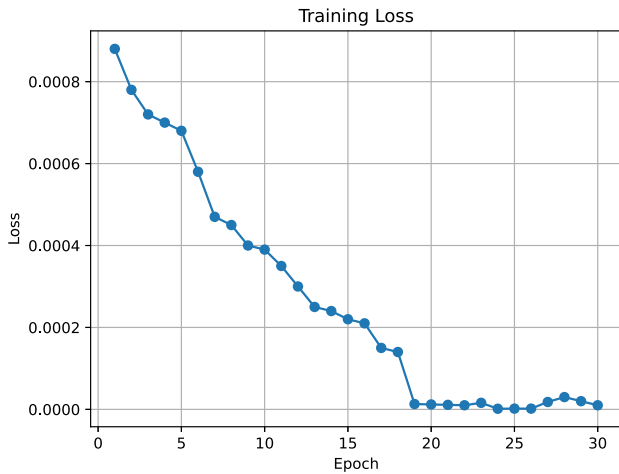


Fig. 4. Training Loss Progression During Model Training (30 Epochs).

TABLE II
PERFORMANCE COMPARISON OF DIFFERENT ANOMALY DETECTION MODELS

Model	Accuracy	Precision	Recall	F1-score
PCA	48.85%	24.42%	50.00%	32.82%
FeatureBagging	49.31%	50.57%	50.21%	40.60%
HBOS	49.56%	51.31%	50.48%	40.67%
Sampling	50.31%	53.36%	51.23%	41.56%
IForest	52.86%	60.53%	53.78%	44.47%
INNE	52.89%	60.57%	53.81%	44.51%
ALAD	53.03%	60.71%	53.95%	44.81%
LOF	53.10%	60.36%	54.00%	45.17%
LODA	53.22%	61.49%	54.15%	44.92%
OCSVM	53.45%	62.02%	54.37%	45.23%
DeepSVDD	53.50%	62.06%	54.42%	45.34%
CBLOF	53.53%	62.24%	54.45%	45.32%
GMM	54.46%	64.87%	55.38%	46.38%
KDE	54.84%	66.10%	55.77%	46.79%
MO_GAAL	55.19%	67.56%	56.12%	47.04%
QMCD	55.60%	67.56%	56.51%	47.86%
VAE	55.81%	68.66%	56.73%	47.96%
AEISVM	55.97%	69.44%	56.90%	48.06%
MCD	57.06%	72.32%	57.99%	49.40%
KNN	57.31%	72.97%	58.24%	49.70%
ABOD	57.46%	73.23%	58.38%	49.90%
COPOD	57.91%	74.78%	58.84%	50.38%
ROD	58.47%	77.01%	59.40%	50.91%
ECOD	58.52%	76.42%	59.45%	51.11%
SOD	58.96%	70.34%	59.79%	53.28%
LUNAR	63.28%	73.71%	64.03%	59.43%
CD	65.84%	69.01%	66.28%	64.72%
AnoGAN	83.12%	84.24%	83.33%	83.04%
autoencoder	95.61%	95.96%	95.61%	95.60%
Proposed	99.11%	100%	98.17%	99.08%

to adapt more effectively to variations in data, ensuring that anomalies are detected more reliably. By leveraging this adaptive threshold, the proposed method achieves better performance compared to other approaches, as it minimizes both false positives and false negatives, making it more robust in real-world anomaly detection tasks.

B. Compare With Supervised Models

In this section, we compare our model with various supervised models. The experiments and results are based on the study by Hussain et al. [43], with details provided in Table III.

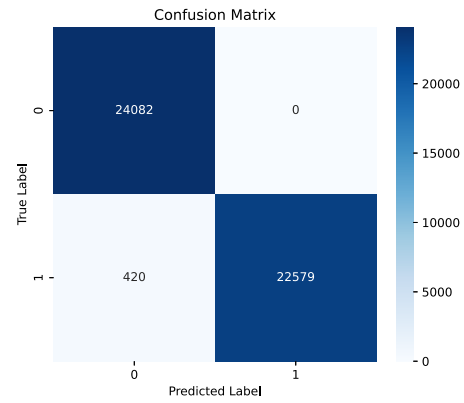


Fig. 5. Confusion matrix for AutoKAN on the testing dataset, showing the number of correct and incorrect predictions for normal and anomaly events.

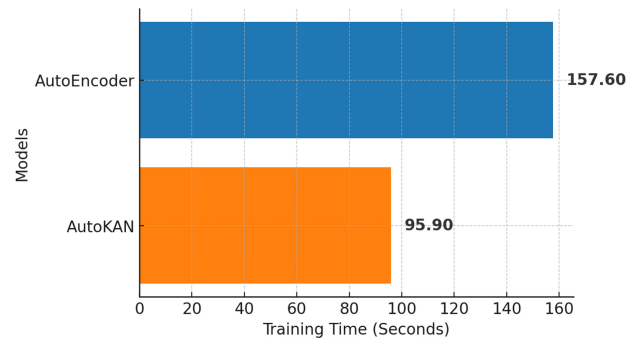


Fig. 6. Comparison of training time between autoEncoder and AutoKAN, demonstrating the improved efficiency of AutoKAN.

Our model achieves the highest precision among all models. However, for other metrics such as recall, accuracy, and F1-score, supervised models demonstrate superior performance. This outcome is expected, as supervised models rely on a large volume of labeled data for training, allowing them to effectively learn and recognize cyberattack patterns.

The key advantage of our model is that it does not require labeled attack data for training. In real-world applications, the lack of labeled data is often a significant challenge, both in terms of data availability and cost. Despite this limitation, our model still achieves comparable performance to supervised models, with only approximately 0.01% error margin, which is an acceptable trade-off given the benefits of unsupervised learning.

C. Model Complexity Compare With Traditional Auto Encoder

In this section, we demonstrate the advantages of AutoKAN compared to the conventional autoencoder in anomaly detection tasks. Additionally, we compare both models in terms of model complexity and runtime performance. Table V provides details on memory usage, FLOPs, and the number of parameters compared between the traditional autoencoder and AutoKAN. The traditional autoencoder has seven times more parameters than AutoKAN while achieving slightly lower performance. Furthermore, AutoKAN requires only 0.01 MB

TABLE III
PERFORMANCE COMPARISON OF ML CLASSIFIERS

Author	ML Classifier	Precision	Recall	Accuracy	F1-score
Hossain et al. [43]	NB	79.67	99.71	52.18	68.51
	KNN	99.65	99.69	99.49	99.59
	RF	99.71	99.80	99.51	99.65
	AB	99.55	99.45	99.50	99.47
	LogR	95.29	90.35	99.50	94.71
	DT	99.69	99.80	99.48	99.64
Our Approach	AutoKAN	100.00	98.17	99.11	99.08

TABLE IV
INFERENCE PERFORMANCE METRICS

Run	Running Time (s)	CPU Usage (%)	Memory Usage (MB)	Traffic per Second
Run 1	0.0440	25.81	1150.86	1068181.82
Run 2	0.0290	25.00	1150.86	1620689.66
Run 3	0.0280	16.67	1150.86	1678571.43
Run 4	0.0280	33.33	1150.86	1678571.43
Run 5	0.0270	25.00	1150.86	1740740.74
Run 6	0.0280	23.61	1150.86	1678571.43
Run 7	0.0270	20.83	1150.86	1740740.74
Run 8	0.0300	20.83	1150.86	1566666.67
Run 9	0.0290	16.67	1150.86	1620689.66
Run 10	0.0270	10.42	1150.86	1740740.74
Average	0.0297	21.82	1150.86	1613416.43

TABLE V
COMPARISON OF AUTOENCODER AND AUTOKAN

	autoEncoder	AutoKAN
Memory Usage (MB)	0.1	0.01
FLOPs (Millions)	1241.43	17.30
Parameters	26807	3359

of memory—ten times less than the traditional autoencoder—while the FLOPs of the autoencoder are nearly eighty times higher than those of AutoKAN.

On the other hand, we recorded the training time in Figure 6, which highlights a significant difference in computational efficiency between the two models. AutoKAN requires only 95.9 seconds to train 50,000 samples, whereas the autoencoder takes 157.6 seconds for the same dataset. This demonstrates that AutoKAN not only accelerates the training process but also scales more efficiently as the dataset size increases. As training data grows, the gap in training time between the two models is expected to widen, making AutoKAN a more practical choice for large-scale applications where computational efficiency is critical.

D. Model Performance in a Resource-Constrained Environment

In this section, we demonstrate the efficiency of AutoKAN in a resource-constrained environment, where computational capacity is highly limited. Table IV provides detailed information based on ten independent runs, reporting the average values. The average runtime per sample is 0.0297 seconds, indicating that AutoKAN can process more than 1.6 million traffic samples per second. This meets the requirements for real-time applications. Even when processing a high volume

of traffic simultaneously, AutoKAN demonstrates remarkable computational efficiency, utilizing only 21.82% of CPU resources and approximately 1050 MB of memory. These findings indicate that AutoKAN can handle substantial workloads without significantly taxing system resources, ensuring smooth and uninterrupted performance. This efficiency is especially vital for deployment in resource-limited environments, including embedded systems, edge computing devices, and IoT applications, where computational power and memory are restricted. AutoKAN stands out as a highly practical solution by minimizing resource consumption while efficiently processing large-scale data in real time. Its ability to perform high-speed anomaly detection with minimal hardware investment makes it well-suited for such scenarios. Its ability to operate effectively under such conditions reinforces its suitability for large-scale, cost-effective implementations across various domains, including cybersecurity, industrial monitoring, and healthcare analytics.

VI. DISCUSSION

A. The Potential of AutoKan in Real-World Application

The AutoKAN model presents a significant potential for real-world deployment in IoT devices for hospitals in monitoring sensor signals. Due to its lightweight design, the model can offer the advantage of cost-effectiveness by using less powerful processors and less memory, which results in a lower price point per device. This is crucial for widespread deployment in health settings, where numerous sensors and monitoring devices might be needed across a hospital or for large numbers of patients. Furthermore, the reduction in device cost makes it more economically feasible to deploy on a large scale. Hospitals can equip more beds and rooms with monitoring capabilities without incurring exorbitant expenses.

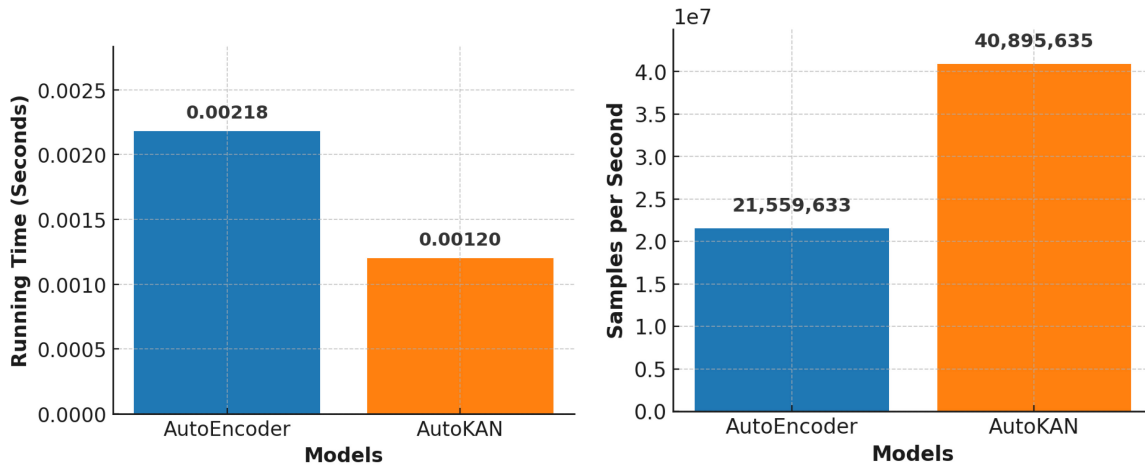


Fig. 7. Comparison of execution time and processing speed for autoEncoder and AutoKAN, highlighting AutoKAN's superior performance.

B. The Ethical and Privacy Implications

Federated learning inherently provides a robust framework for data privacy. In our framework, the federated averaging algorithm is utilized to ensure that patient data remains decentralized. Instead of transmitting raw data to a central server for AI model training, federated learning offers a decentralized solution where each local device trains its local AutoKAN model using its own data. While KANs offer better explainability than traditional MLPs, the federated learning process itself can introduce complexity. Ensuring transparency in the model aggregation process and providing explanations for the model's decisions are crucial ethical considerations, particularly in healthcare. Furthermore, establishing clear data governance and responsibility locally is another essential ethical and privacy aspect. Due to the model's local deployment, it becomes more sensitive to adversarial attacks, which also raises ethical concerns about the reliability and security of the AI system in patient care.

C. Limitation and Future Work

Through our experiments and results, we demonstrate the potential of AutoKAN as a smart anomaly detection device with the capability for large-scale deployment at a reasonable cost. However, all testing in this study was conducted through simulations, lacking real-device evaluations to validate its performance in practical applications. In the future, we plan to develop a smart device for anomaly detection using sensor signals to monitor diabetes patients in hospitals. Furthermore, evaluating the proposed approach on a larger and more diverse scale, encompassing approximately 50 to 100 devices deployed across multiple hospital wards, particularly in challenging non-IID scenarios with varying patient demographics and sensor noise profiles, would provide a more robust demonstration of its performance and generalizability in real-life applications. This expanded evaluation would also allow us to assess the system's scalability and its ability to handle the complexities of a real hospital environment, including network latency and data synchronization issues. One key reason we chose Kolmogorov–Arnold Networks (KAN) over traditional black-box neural networks is their inherent explainability. As a next step, we aim

to further explore and refine explainable AI (XAI) approaches to interpret our model's decision-making process with greater granularity. This will involve not only identifying which input features are most influential in the model's predictions but also quantifying the uncertainty associated with those predictions and providing clinicians with actionable insights into the factors contributing to each anomaly detection. This enhanced explainability will help build trust and facilitate adoption among users, such as hospitals and IT administrators, ensuring transparency, reliability, and clinical utility in real-world applications. Additionally, the AutoKAN federated learning model is indeed designed to preserve data privacy by ensuring patient data remains decentralized and is not transmitted to a central server. This fundamental architectural choice inherently supports privacy preservation, which is a key tenet of regulations like HIPAA and GDPR. While the current study lays this strong privacy-preserving foundation, future work will focus on explicitly demonstrating compliance with such stringent healthcare regulations.

VII. CONCLUSION

In this study, we introduce AutoKAN, an anomaly detection model integrated with a federated learning mechanism. AutoKAN outperforms over 28 state-of-the-art anomaly detection models, achieving an impressive 99% accuracy and 100% precision. Additionally, it demonstrates its capability to function efficiently in resource-constrained environments with limited computational resources. By integrating Kolmogorov–Arnold Networks (KAN) into the autoencoder architecture, AutoKAN serves as a lightweight and faster alternative to traditional neural networks. Additionally, the adaptive threshold strategy has shown its effectiveness through experimental results. In conclusion, the proposed framework holds significant application value in medical anomaly detection and has the potential to be extended to other fields requiring efficient and accurate anomaly detection.

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